

ClaimSense 360: Master Viva & Examiner Defense Handbook

COMPLETE 26-MODULE ANSWERS, MODEL JUSTIFICATIONS & PRESENTATION SCRIPTS

5-Minute Presentation Script: 'Respected Sir/Examiners, today I present **ClaimSense 360**, an Enterprise AI Motor Insurance Claims Intelligence Platform. Motor insurance fraud costs Indian insurers over **₹45,000 Crore** annually. Legacy rule-based software fails because fraud syndicates bypass static limits, while black-box AI is illegal under IRDAI laws because it lacks mathematical proof. ClaimSense 360 solves this using a Multi-Modal AI Engine: **XGBoost** computes fraud probability, **SHAP Game Theory** provides mathematical proof, **TF-IDF NLP** detects written deception, **YOLOv8 + PyTorch ResNet-18** measures damage severity, and **Digital Image Forensics** catches Google-downloaded stock photos via EXIF telemetry.'

1. Project Rationale & Problem Statement

Problem: ₹45,000 Crore annual loss in India due to staged claims and recycled web photos.
Solution: Sub-second multi-modal AI providing mathematical SHAP proof and EXIF forensics.

2. Frontend Architecture & Design System

Built with Next.js 15.1 App Router + React 19 + TypeScript. Styled using HSL Design tokens (#173B32 Deep Forest Green, #E66A4E Terracotta Orange, #C9FF3D Lime Accent) and Recharts 2.15.

3. Authentication & User Roles

PyJWT OAuth2 Bearer Tokens + bcrypt 4.0.1 password hashing + HTTP-only cookies. Supports Customer, SIU Investigator, Claims Manager, and Admin roles.

4. AI/ML Models & Mathematical Proofs

XGBoost Fraud Model: Trained on 1,000 Kaggle claims across 46 features.

SHAP Game Theory Equation: $\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \left[\frac{|S|!(|N|-|S|-1)!}{|N|!} \right] * [v(S \cup \{i\}) - v(S)]$

Computer Vision Damage Score: $Severity = 0.50 * S_{YOLO} + 30 * C_{ResNet} + 20 * E_{Edge}$

EXIF Image Forensics: Detects missing camera headers on web downloads, adding +1.850 to SHAP and elevating risk score to High Risk (70.3/100).

AI Model	Technology	Primary Role	Math / Method
Tabular Fraud Classifier	XGBoost 2.0	Fraud Probability Scoring	Gradient Boosted Trees
Explainable AI	SHAP TreeExplainer	Marginal Feature Attribution	Shapley Game Theory
Computer Vision	YOLOv8 + PyTorch ResNet-18	Vehicle Damage Severity	22-Class Detection + L2 Norm
NLP Deception Engine	TF-IDF + Ridge	Narrative Suspicion Score	1-gram/2-gram Cosine Distance
Image Forensics	PIL EXIF Telemetry	Web/Stock Photo Detection	Raw Camera Header Scanning

5. Critical Examiner Defense Q&A;

Examiner: 'Is this AI actual or hardcoded/dummy?'

Your Defense Answer: 'Sir, our AI engine is 100% real and verifiable in real-time. We have live trained fraud_model.joblib (XGBoost), shap.TreeExplainer, yolov8n.pt PyTorch weights, and PIL EXIF telemetry. When you submit a new claim or upload a downloaded image, the model computes real-time SHAP factor contributions and renders a live Image Forensics Warning badge. We can run our Python test verification script verify_yolo_execution.py right now to prove live GPU tensor execution!'